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Deep eutectic solvent-infused two-dimensional metal–organic framework membranes as quasi-solid-state electrolytes for wearable micro-supercapacitors†

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The burgeoning field of miniaturized and portable electronic devices calls for novel advances in micro-energy storage technology. Micro-supercapacitors (MSC) stand at the forefront of this endeavour, yet unlocking their full potential necessitates the exploration of high-performance electrolytes. Herein, we introduce a strategy that leverages flexible metal–organic framework (MOF, CuTCPP) nanosheet-based membranes to construct quasi-solid-state electrolytes (QSSEs) and enhance the ionic conductivity and electrochemical performance of deep eutectic solvent (DES)-based MSCs. Owing to the multiple nano-channel pathways provided by the porous MOF nanosheets, the ionic conductivity of DES within the nanochannels exhibits a 13-fold increment compared with its bulk counterpart. Furthermore, we engineered MSC harnessing the CuTCPP–DES system, whose performance surpasses that reported for most of the ionic liquid and 2D material-based MSCs. The areal-specific capacitance was 81.3 mF cm^{-2} at a current density of 0.1 mA cm^{-2} , and the energy density was $45.17 \text{ μW h cm}^{-2}$ at a power density of 8.559 mW cm^{-2} . Notably, the performance of MSCs remains consistent and unaffected, even when subjected to bending. These findings contribute to the exploration and potential optimization of the inherent benefits of MOFs, thereby presenting a paradigm shift in nanoconfined systems for microscale energy storage applications.

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Introduction

The rapid advancement of wearable and flexible electronic devices has spurred a concurrent increase in demand for compact, high-capacity, and flexible next-generation energy storage systems, notably micro-supercapacitors (MSCs). As a core unit within MSCs, electrolytes are expected to be the nexus of decent ion conductance, reliable operation safety, long cycle life, and low-cost yet scalable production. The conventional liquid-state electrolytes, mostly organic-based, are prone to leaking during use, causing severe safety concerns.

Instead, all solid-state electrolytes with leak-free features are currently the dominating power of MSCs. However, despite their prevalent use, these electrolytes confront numerous challenges, including low ionic conductivity, high interface resistance, and limited compatibility with various materials used in energy storage systems. These limitations compromise device performance and constrain their operational capabilities.^{1–6}

Combining the advantageous properties, quasi-solid-state electrolytes (QSSEs) have emerged as promising alternatives. This system integrates a solid matrix encapsulated with liquid electrolytes, and the mixed structure offers the mechanical stability of solid electrolytes (safety) while preserving the high ionic conductivity inherent to liquid electrolytes (performance). Moreover, the inherent malleability of QSSEs also promotes superior contact with solid device components, decreasing interface resistance and enhancing overall device performance.^{2,7–10} To enable QSSEs as a promising frontier in the quest for next-generation electrolytes, its design principle is to build a large specific surface area and a framework that absorbs electrolytes. 2D materials, having a layered structure

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and broad layered transport channels, have become promising candidates for designing platforms for QSSE. However, their mass transfer resistance is relatively high due to the long ion transport pathway. Therefore, it is particularly important to reduce the mass transfer resistance of electrolytes in 2D materials.

Within this context, metal–organic framework (MOF) nanosheets have emerged as potential game-changers in energy conversion and storage owing to their customizable structures, ordered pores, large surface areas, and accessible side groups. MOF nanosheet-based membranes have particularly displayed immense potential as QSSE platforms owing to their unique characteristics. Their porous structures facilitate efficient liquid storage, while their nanoscale confinement promotes enhanced ionic conductivity. Additionally, their nanosheet morphology and porous nature enable multiple nanochannel pathways, and the abundance of functional groups available for modification allows for performance designation. Recently, MOFs have been used in the electrochemical field for achieving high-performance devices.^{11–14} However, these MOF nanosheets face challenges of their own, including their preparation being complex, they are typically unstable, and their rigidity limiting their application in energy storage.^{15–19} To address these issues, we have focused on a flexible MOF known as copper tetrakis (4-carboxyphenyl) porphyrin (CuTCPP), which can be easily prepared into ultrathin nanosheets and reassembled into thin films, thereby offering abundant interstitial nanochannels for ion transport. Concurrently, deep eutectic solvents (DESs), a class of ionic liquid-like salts composed of hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA), show notable promise as QSSE for wearable MSCs. They have garnered significant attention owing to their low cost, ease of preparation, non-toxicity, biodegradability, and non-flammability. Furthermore, the properties of DESs can be precisely tuned by adjusting the type and ratio of their components, thus specifically tailorable for wearable devices.^{20–26}

To consolidate these advances, we present a flexible CuTCPP–DES membrane-based QSSE to delve into the mechanism behind significantly enhanced ionic conductivity and overall MSC performance. The ionic conductivity of nanoconfined DES surpasses that of the bulk counterparts by a factor of 13 times, which is attributed to the multiple pathways provided by porous MOF nanosheets and nanoconfinement. Practical applications of nanoconfined DES MSCs reveal superior electrochemical performance and mechanical flexibility. Specifically, the best CuTCPP–DES MSC demonstrated an exceptional areal-specific capacitance of 81.3 mF cm^{-2} at 0.1 mA cm^{-2} and an energy density of $45.17 \text{ } \mu\text{W h cm}^{-2}$ at a power density of 8.559 mW cm^{-2} . Importantly, the performance remains impervious to bending, indicating its robust and versatile applicability. This approach holds the promise of wide-ranging applications, especially in electrochemical energy storage for wearable MSCs, thus offering a practical, high-performance solution to the current challenges of energy storage in wearable electronics.

Methods

Chemicals and materials

Choline chloride (AR) and 1,2-butanediol (99.9%) were purchased from Aladdin. Copper nitrate trihydrate $[\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}]$, trifluoroacetic acid (TFA), and tetrakis-(4-carboxyphenyl)-porphyrin (TCPP) were bought from Sigma-Aldrich. Polyvinyl pyrrolidone (PVP M.W. 58 K) and dimethyl formamide (DMF, AR) were purchased from Merck. Ultrapure water of conductivity $18.2 \text{ M}\Omega \text{ cm}^{-1}$ was produced by a Millipore direct-Q system (Millipore, U.S.A). Activated carbon (AC) and carbon sheets were purchased from Nanjing Xianfeng Nano Material Technology Co., Ltd. Carbon black, poly (vinylidene fluoride) (PVDF), and *N*-methylpyrrolidone (NMP) were purchased from Aladdin. All reagents and solvents were used as received.

Synthesis of 2D CuTCPP nanosheets

CuTCPP nanosheets were synthesized by the hydrothermal method as previously reported.²⁷ First, 0.03 mM copper nitrate trihydrate and 20 mg PVP were placed in the Teflon container with the addition of 24 mL mixed solution of DMF and ethanol (3 : 1) and 20 mL TFA solution for ultrasonic treatment for about 20 minutes. 8 mL of the mixed solution of DMF and ethanol (3 : 1) was added to 8 mg TCPP and dispersed by ultrasonication for 20 min. The dispersed TCPP solution was dripped into the container of the reactor one by one and then reacted for 3 hours at 80°C in the oven. After the reaction, the product was centrifuged with ethanol at $12\,000 \text{ rpm min}^{-1}$ for three times. Finally, CuTCPP nanosheets dispersed in ethanol were obtained.

Synthesis of the choline chloride-1,2-butanediol DES solution

Choline chloride (ChCl) and 1,2-butanediol (BD) with specific molar ratios were added into sealed beakers. In the pre-experiment, it was found that the DES would precipitate under ambient conditions with the ChCl : BD ratios of 1 : 1 and 1 : 2. Therefore, ChCl : BD ratios of 1 : 4, 1 : 6, 1 : 8, and 1 : 10 were chosen to form a homogeneous and stable system. The mixtures were heated at 70°C for about an hour until a transparent and stable liquid solution was formed. Among them, the molar ratios of ChCl : BD = 1 : 4, 1 : 6, 1 : 8, and 1 : 10 were named DES4, DES6, DES8, and DES10, respectively.

Synthesis of CuTCPP–DES membrane and nanofluidic devices

3 mL of the as-prepared CuTCPP nanosheet dispersion was used to fabricate the CuTCPP membrane on anodic alumina oxide (AAO) substrates by vacuum filtration. The CuTCPP membrane can be easily peeled off from AAO to obtain self-standing membranes. These membranes were then dried overnight for use in the subsequent experiments. After dropping and immersing 50 μL DES solution on the CuTCPP membrane for 30 min, the membrane was spin-coated at 6000 rpm for 60 s to obtain the CuTCPP–DES membrane. The CuTCPP–DES membrane was cut into rectangular pieces and immersed in a blend of a polydimethylsiloxane (PDMS) precursor and curing

agent (20:1). After curing, two reservoirs were carved out in the PDMS device to expose the two ends of the membrane to store the ChCl-BD electrolyte.

Fabrication of a coin cell supercapacitor and a micro supercapacitor

Typically, the activated carbon sample (80 wt%), conductive carbon black (10 wt%), and polytetrafluoroethylene (10 wt%) were mixed in ethanol and magnetically stirred for 24 h. Then, the mixed slurry (active carbon materials of 1–2 mg) was uniformly coated onto a carbon sheet ($\sim 1 \text{ cm}^2$) and dried at 120°C in an oven for 12 h to prepare the working/counter electrode (maintaining the mass of carbon material on the two electrodes in the same cell is same). The CuTCPP dispersion was vacuum filtered on the glass fiber and then dried in the oven at 120°C for 4 h. Then, the DES solutions were dropped and spin-coated on the CuTCPP membrane to prepare the separator of the CuTCPP-DES supercapacitor. Subsequently, we produced sandwich-type supercapacitor coin cells consisting of a CR2032 button cap, two carbon electrodes, a CuTCPP-DES membrane, and a top cap. Each component was placed sequentially in a hydraulic crimper (MSK-110, MTI Corporation, Richmond, CA, USA) and sealed at a pressure of 50 kg cm^{-2} . For the micro supercapacitor, to fabricate the interdigital AC electrode, a customized interdigital mask with three fingers (length of 5.0 mm, finger width of 1.0 mm) on each side was manufactured by laser direct writing to define the patterned geometry. The nanoconfined DES electrolytes were prepared by immersing the CuTCPP membrane in the DES solution followed by spin-coating. Two AC interdigital electrodes were placed on the nanoconfined DES to assemble a nanoconfined DES MSC. Polyethylene terephthalate (PET) adhesive tape was used to seal the MSC.

Characterization

Field-emission scanning electron microscopy (FE-SEM) and energy-dispersive X-ray spectrometry (EDS) was employed to obtain SEM and EDS mapping images of CuTCPP and CuTCPP-DES using a Sigma 300 SEM instrument equipped with an EDS spectrometer. A Shimadzu XRD-6000 with a Cu K α radiation source was used to obtain X-ray diffraction (XRD) patterns of CuTCPP and CuTCPP-DES. A contact angle was measured using a Data-Physics OCA-20 video-based contact angle measuring device. Fourier transform infrared (FTIR) reflection spectra of the liquid and film samples were measured using a Nicolet 5700 infrared spectrometer, Thermoelectric Company of America. The spectra were used to analyze the surface functional groups of the liquid solvents and film samples. Atomic force microscopy (AFM) (Bruker Dimension Edge 8) was used to characterize the thickness and morphology of the nanosheets in tapping mode. EDS analysis and transmission electron microscopy (TEM) (HT7700) were employed to assess nanosheet and membrane morphology. The zeta potential of the nanosheets was obtained using a zeta potential analyzer, (Zetasizer Nano-ZS, Malvern, UK). X-ray photoluminescence spectroscopy (XPS) spectra of the CuTCPP

membrane were recorded using an XPS AXIS SUPRA (Kratos, UK) instrument.

Electrochemical characterization

The ion-transport behavior of bulk and nanoconfined DES systems was studied using EIS of nanofluidic devices tested at the electrochemical workstation (CHI-660D, Shanghai ChenHua Instruments Co., Ltd, China). EIS plots were obtained from 0.01 Hz to 100 kHz. The Nyquist diagram reveals the ion transport of electrolytes between electrodes, where the charge transfer resistance can be determined by the semicircle diameter (R_{ct}). The intersection point refers to the equivalent series resistance (R_s), including the resistance of the electrode, the connection resistance, and the electrolyte of the device. The diameter of a semicircle is related to the charge transfer resistance, which includes the electrode-electrolyte interface and the pore structure of the electrode. Z-view software was used to obtain the resistance values of the measured electrochemical systems. Cyclic linear scan voltammetry (LSV) was performed to measure the electrochemical stability window of the DES electrolyte at 10 mV s^{-1} (Fig. S1†). Specific areal and volumetric capacitances of supercapacitors were determined using CV curves at the electrochemical workstation. The cycling performance, galvanostatic charging-discharging (GCD), and charge-discharge curves were measured using a Land CT2001 A instrument. CV tests were performed with the potential window from 0 V to 2.0 V at scan rates varying from 50 mV s^{-1} to 250 mV s^{-1} . The GCD tests were performed at different current densities ranging from 0.1 mA cm^{-2} to 2 mA cm^{-2} .

A strip of CuTCPP film was embedded in a PDMS elastomer with two reservoirs containing approximately 0.1 mL of DES volume. Two Pt wires were used as electrodes to explore the electrochemical properties. The ionic conductivity of the nanoconfined DES can be calculated using eqn (1):

$$\sigma = \frac{l}{R \times S} \quad (1)$$

where σ is the ionic conductivity, l is the distance between the two Pt wires, R is the resistance, which can be obtained from the Nyquist plots, and S is the contact area.

The capacitance of the supercapacitor is calculated using eqn (2):

$$C = \frac{I \times \Delta t}{\Delta V} \quad (2)$$

where I is the charge/discharge current, Δt is the discharge time, and ΔV is the voltage window. The specific areal energy density and power density were defined according to eqn (3) and (4), respectively:

$$E = \frac{C \times \Delta V^2}{7200} \quad (3)$$

$$P = \frac{E \times 3600}{t} \quad (4)$$

where E is the energy density ($\mu\text{W h cm}^{-2}$), and P is the power density ($\mu\text{W cm}^{-2}$).

Results and discussion

Membrane preparation and characterization

CuTCPP nanosheets synthesized by Cu^{2+} ions and porphyrin ligands were well dispersed into a stable and translucent purple colloidal solution due to their negatively charged surface (Fig. 1a and Fig. S2† for Tyndall effect and zeta potential). The TEM image captured the transparent and wrinkled surfaces of these nanosheets, indicating their ultrathin and flexible nature (Fig. 1b). Atomic force microscopy (AFM) analysis confirmed the large aspect ratio of the CuTCPP nanosheets with a lateral size of 1 to 1.2 μm and a thickness of approximately 4 nm (Fig. 1c). While dispersed CuTCPP nanosheets possessed a regular 1.19 nm pores (Fig. S3†),²⁸ such a size could be reduced to approximately 5.4 Å due to their AB stacking mode along the c -axis in the assembly, as corroborated by a few previous studies.^{29–31} After vacuum filtration, the nanosheets could be self-assembled and processed into a freestanding membrane with a defect-free surface and layer-like cross-section (Fig. 1d). The as-prepared MOF membrane preserved the nanosheets' flexibility and could be folded

and bent at will, which is a critical prerequisite for its real-world application as wearable electronic units (Fig. 1e).³²

The DES employed in this study (Fig. S4 and Table S1†) consists of H-bond donor choline chloride (ChCl) and H-bond acceptor 1,2-butanediol (BD) with varied ChCl:BD molar ratios (1:4, 1:6, 1:8, and 1:10, denoted as DES4, DES6, DES8, and DES10, respectively). The excellent wettability on the MOF membrane (Fig. 1e) allowed its rapid infusion into the interlayer space between the CuTCPP layers, increasing the membrane thickness from 1.6 μm to 2.8 μm (Fig. 1d and Fig. S5†). This change was in accordance with the interlayer spacing (d -spacing) enlargement from 0.459 nm to 0.484 nm, as reflected by the shift of (004) peak from 19.3° to 18.3° in the X-ray diffraction (XRD) pattern (Fig. 1f). The successful integration of DES into CuTCPP was also confirmed from the change in the chemical composition of the membrane. Without the DES treatment, CuTCPP displayed two characteristic peaks at 1616 cm^{-1} and 1402 cm^{-1} in the FTIR spectrum, attributed to $\text{O}=\text{C}-\text{O}-\text{Cu}$ bonds following $\text{Cu}^{2+}-\text{COOH}$ (porphyrin) coordination (Fig. 1g).³³ After DES impregnation, a major new peak at 3373 cm^{-1} , intrinsic to the stretching vibration of $-\text{OH}$ groups in DES, was observed for the CuTCPP–DES hybrid system. Moreover, X-ray photoelectron spectroscopy (XPS) of the CuTCPP–DES also confirmed the presence of additional Cl from ChCl in DES-impregnated CuTCPP (Fig. 1h, Tables S2 and S3†). Collectively, these find-

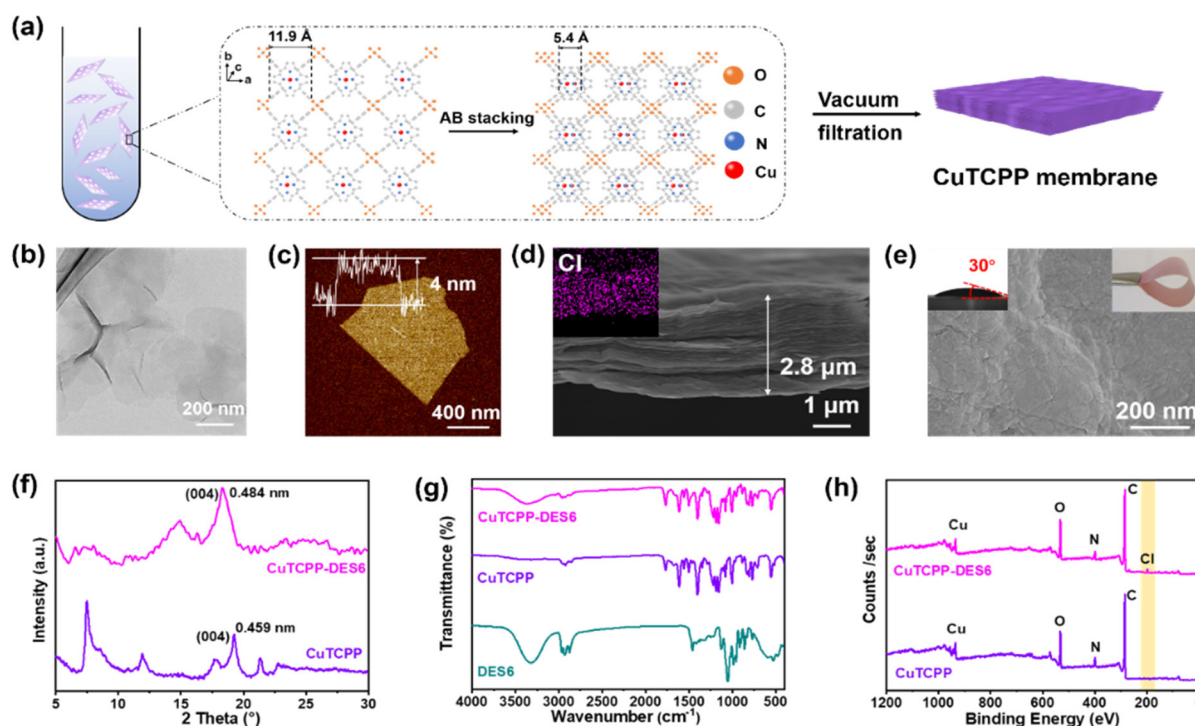


Fig. 1 (a) Schematic diagram of the narrow pore size of the CuTCPP membrane compared with a single CuTCPP layer. (b) TEM image of CuTCPP nanosheets. (c) AFM image with the corresponding thickness profile of CuTCPP nanosheets. (d) Cross-sectional SEM image of the CuTCPP–DES membrane, inserted image is the EDS mapping image of the Cl element. (e) Surface SEM image of the CuTCPP–DES membrane. The inserted image shows the contact angle of DES on the Cu–TCPP membrane and the digital photo of the flexible CuTCPP membrane. (f) XRD patterns of the CuTCPP and CuTCPP–DES6 membranes. (g) FTIR spectra of DES6, CuTCPP, and CuTCPP–DES6. (h) X-ray photoelectron spectroscopy (XPS) spectra of the CuTCPP and CuTCPP–DES6 membranes.

ings affirmed the successful generation of ultra-thin, uniform CuTCPP nanosheets and the penetration of DES into the inter-layer of the nanosheets.

Ionic conductivity behavior of the nanoconfined DES

The ion transport behavior in the CuTCPP–DES membrane was studied by monitoring its ionic conductivity in a home-made device using electrochemical impedance spectroscopy (EIS) studies (Fig. 2a). Prior to the measurements, the ionic conductivities of the DES bulk solution with different ChCl:BD ratios were screened to ascertain the most appropriate testing conditions (Fig. S6†). At 20 °C, it was found that DES was the most conductive (2.03 mS cm^{-1}) when the ratio reached 1 : 6 (DES6), which was because of the balanced hydrogen bond (H-bond) between its donor and acceptor. Below this

ratio, the increase in butanediol could weaken the H-bond between the pair, leading to more complete ion dissociation and thus promoting ion transport. Furthermore, increasing the butanediol content beyond this ratio, however, would disturb the equilibrium and the redundant butanediol would cause steric hindrance that unfavored ion transport. As shown in Fig. S7,† the viscosity of DES decreases with the increase of butanediol, and DES6 exhibited the least viscosity, which contributes to ion transport. With the addition of choline chloride in DES, the increased viscosity can be mostly attributed to the formation of hydrogen bonds that result in the loss of molecular mobility. Meanwhile, previous simulation³⁴ and self-diffusion coefficient tests (Fig. S8†) proved that DES6 exhibited the best conductivity performance under nanoconfinement. Therefore, DES6 was chosen as the media for the following

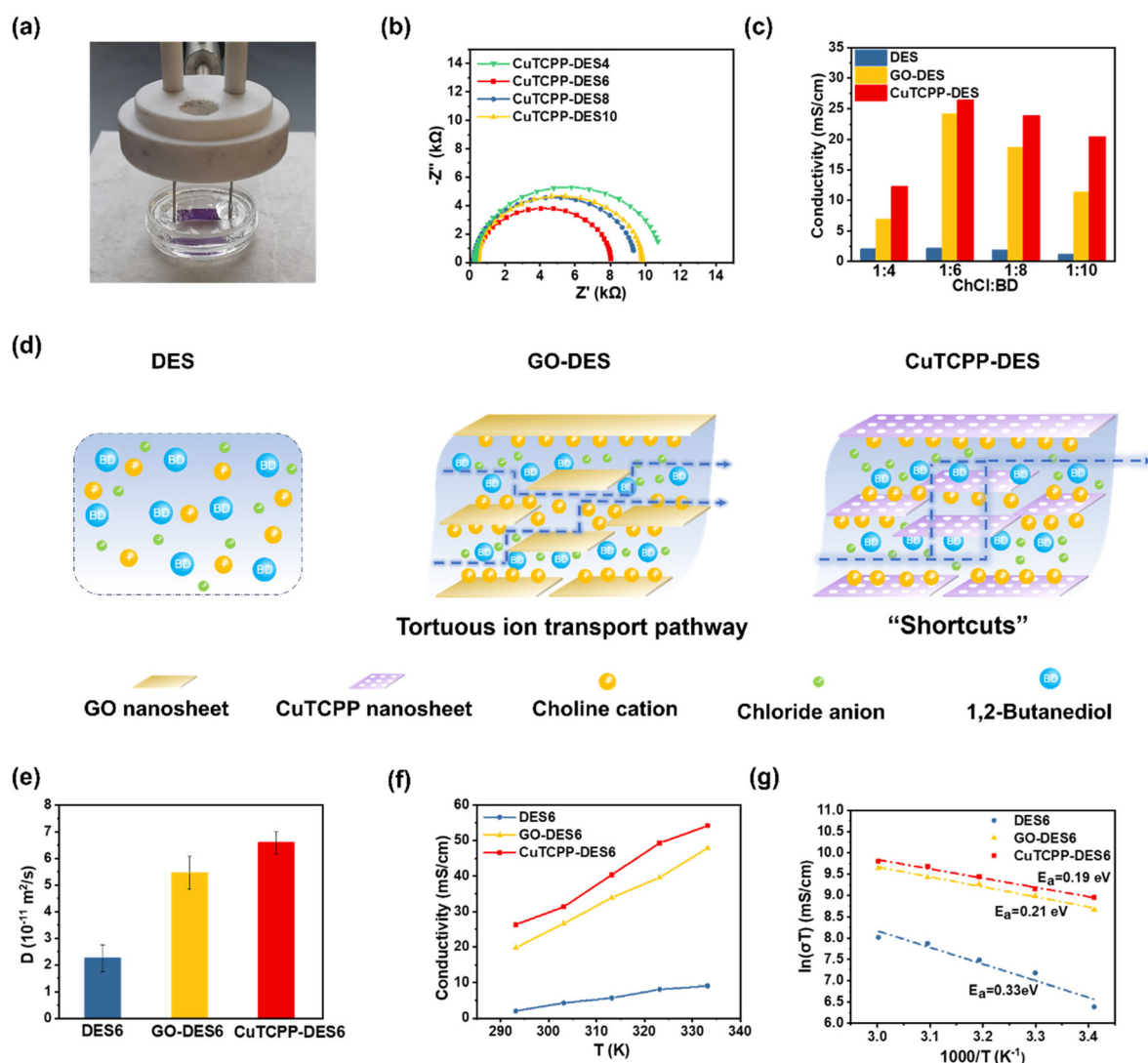


Fig. 2 (a) The nanofluidic device used for the electrochemical measurement. (b) Nyquist plots of nanoconfined DES systems. (c) The increment of the conductivity of the DES is due to the nano-confinement of GO and CuTCPP at ambient conditions. (d) Schematic illustration of DES, GO–DES, and CuTCPP–DES. (e) Self-diffusion coefficients of DES6, GO–DES6, and CuTCPP–DES6. (f) The ionic conductivity of bulk and nanoconfined systems against temperature. (g) Arrhenius plots of the conductance ($\ln(\sigma T)$) vs. inverse temperature ($1000/T$) for DES6, GO–DES6, and CuTCPP–DES6.

investigations and comparisons. Similar to the bulk phase, DES6 exhibited the highest ion conductivity in CuTCPP-DES hybrid systems with a 13-time enhancement from its bulk counterpart to 26.32 mS cm^{-1} . Such an increase was also observed for the other systems incorporating DES4, DES8, and DES10 (Fig. 2b). This substantial discrepancy can be largely attributed to the nanoconfinement-induced ion stratification phenomenon, as we proved in our previous studies.¹⁰ In 1 nm nanochannels, the charged nanosheet surface will significantly attract counter ions while pulling away co-ions, thus forming two separate ion layers in its close vicinity and in the middle of the channel. The stratification reduces ion dragging and crowding and allows more rapid ion diffusion throughout the nanochannel.^{35,36}

To demonstrate the advantage of using porous MOF membranes as potential electrolytes, we compared their ion-conducting performance with that of graphene oxide (GO) membrane-based system, which is often used as a classical benchmark material. It can be clearly seen that GO-DES6 also demonstrated greatly enhanced conductivity (24.17 mS cm^{-1}), further confirming the promotive role of nanoconfinement in ion transport (Fig. 2c). However, this value was lower than that of CuTCPP-DES6 under the same testing conditions. We attribute the superior ion-conducting performance of CuTCPP membranes to the intrinsic pores on MOF building blocks. The porosity of a material can substantially enhance its surface area, thereby boosting ion transport. This enlarged surface area amplifies the contact region between the ions and the material, increasing the ion adsorption and desorption rates.³⁷ While primary charge transfer may be limited to the small electrode-electrolyte interface, the rest of the electrolyte layer still stores and transmits ions. Thus, transmembrane ion conduction within the electrolyte layer notably impacts device performance.³⁸ Mass transport through the 2D laminar membranes is considered to mostly take place in the horizontal channels between the neighboring layers, therefore, in the GO-DES system, ions will have to take numerous turns and twist to exit the membrane (Fig. 2d). However, in the CuTCPP-DES system, the regular porous structure with proper size ($5.4 \text{ \AA}$) provides many vertical “shortcuts” for ions to shuttle between membrane layers. As a result, the total transporting path length is theoretically reduced to improve the overall ion transporting efficiency. The additional vertically-transporting length L can be calculated using eqn (5).³⁹

$$L = \frac{h}{d} l \quad (5)$$

where h is the transmembrane transport path, d is the inter-layer distance, and all nanosheets are assumed to have equal length l . The ratio of h to d indicates the number of turns a fluid must navigate and the individual sheets, it traverses along the pathway.

Rather than following lengthy and convoluted routes, ions tend to prefer shortcuts through the pores, allowing them to quickly reach the next layer. In the CuTCPP-DES system, the pores on the MOF nanosheets decreased the overall length L .

This straightforward flowing mode cross layers substantially decreases the pathway length L and facilitates mass transport to a great extent.

As a reliable tool to probe the self-diffusing property, pulsed-field gradient nuclear magnetic resonance (PFG NMR, Fig. 2e and Fig. S9†) was employed to corroborate the above discrepancy between the three systems. Within expectation, CuTCPP-DES6 exhibited the largest self-diffusion coefficient of $6.58 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, followed by GO-DES6 with a coefficient of $5.46 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$. These values were 2 to 3 times higher than that of bulk DES6 ($2.25 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$), indicating the significance of smartly engineered nanochannels in high-performance membrane-based QSSEs.^{40–44}

Considering the wide temperature range in which QSSEs may work, we studied the ion conductivity of the three systems between 20 to 60 °C in a constant temperature box. Across the tested range, the primary conductivity sequence of CuTCPP-DES6 > GO-DES6 > DES6 remained unchanged (Fig. 2f). Nevertheless, unlike the DES6 bulk, which illustrated a minor heat response and only had its conductivity raised by 7.04 mS cm^{-1} , CuTCPP-DES6 experienced a conductivity surge by 27.87 mS cm^{-1} . This not only testified to the stable performance of CuTCPP-DES6 as a QSSE but also promises its real-world use for energy storage applications, even under harsh environments. The temperature-conductivity relationship also fits into the Arrhenius formula and allowed us to further probe the ion transport mechanism. By plotting the relation, we managed to calculate the activation energies (E_a) of DES6, GO-DES6, and CuTCPP-DES6 as 0.33 eV, 0.21 eV, and 0.19 eV, respectively (Fig. 2g). The low E_a suggested that protons were transported in the Grotthuss mode,⁴⁵ and the lowest E_a of CuTCPP-DES6 further proved the minimal transporting energy barrier in this system due to the porous structure.

Finally, as a proof of concept, coin cells utilizing CuTCPP-DES6 and GO-DES6 as QSSEs and separators were fabricated to test the performance of vertical transmembrane transport of ions. In this case, the electrolyte is transferred in a horizontal direction. The capacitances of DES6, GO-DES6, and CuTCPP-DES6 were 90 F g^{-1} , 125.5 F g^{-1} , and 141.6 F g^{-1} , respectively, with CuTCPP providing more ion transport channels, thereby enhancing the conductivity and coin cell capacitance of the system (Fig. S10†).

Fabrication and electrochemical performance of CuTCPP-DES MSC

The fabrication process of the CuTCPP-DES MSC is shown in Fig. S11,† resulting in a device that embodies the qualities of both flexibility and wearability (Fig. 3a–c). SEM characterization underscored the interdigital electrode gap of approximately 0.2 mm, with EDS mapping further confirming the placement of the CuTCPP-DES membrane under the microelectrode (Fig. 3d). The electrochemical performance of the CuTCPP-DES MSCs was probed using a combination of cyclic voltammetry (CV) measurements at scan rates varying from 50 to 250 mV s^{-1} and galvanostatic charging/discharging (GCD) tests at current densities between 0.1 and 2 mA cm^{-2} , at a

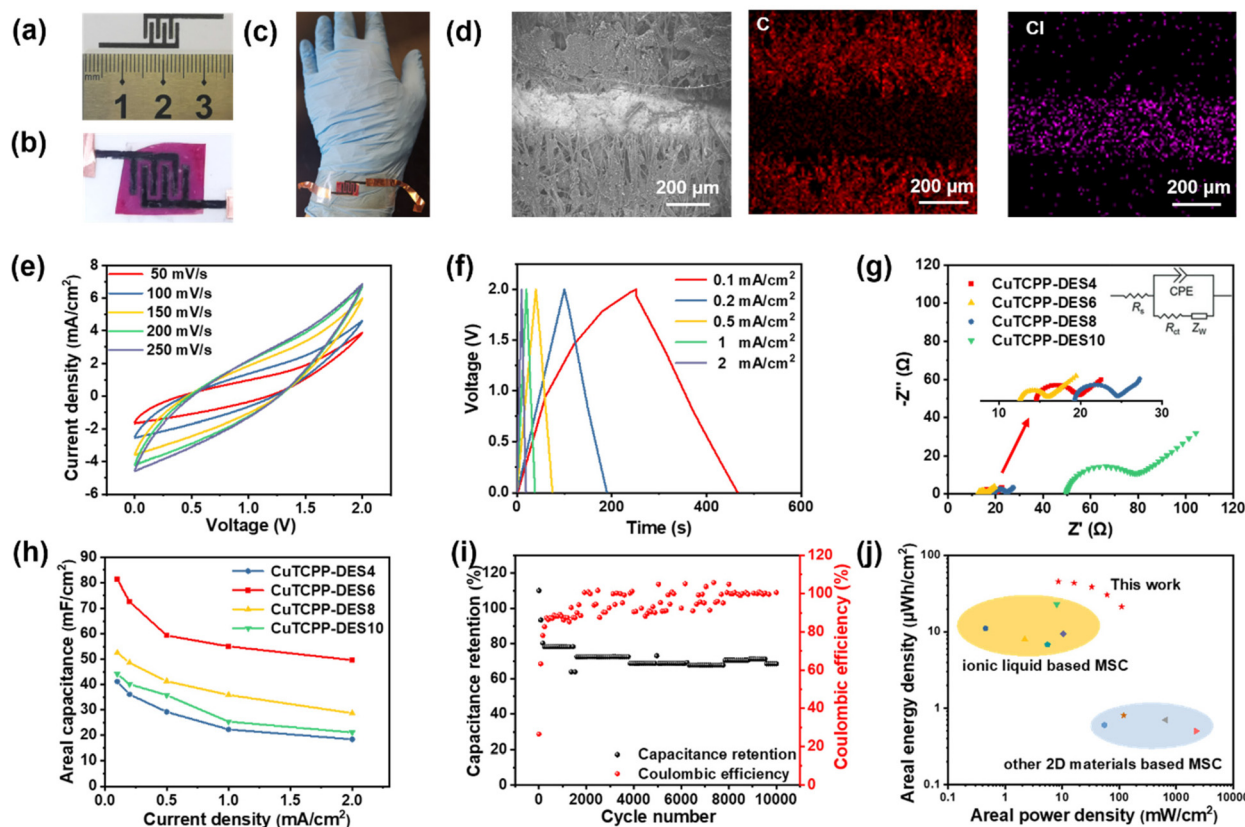


Fig. 3 (a) The digital photo of interdigital AC electrodes. (b and c) The digital photo of a single flexible and wearable CuTCPP-DES MSC. (d) SEM and EDS images of CuTCPP-DES6 MSC surface. (e) CV curves of CuTCPP-DES6 MSC at different scanning rates. (f) GCD curves of CuTCPP-DES6 MSC at different current densities. (g) Nyquist plots of CuTCPP-DES MSCs, inserted image is the equivalent circuit and magnified plots of CuTCPP-DES4, CuTCPP-DES6, and CuTCPP-DES8. (h) Areal capacitance vs. current density of CuTCPP-DES MSCs. (i) Cyclic charge-discharge tests of capacitance retention and coulombic efficiency of CuTCPP-DES6 MSC system at the current density of 0.2 mA cm^{-2} . (j) Ragone plot of volumetric energy density versus power density of CuTCPP-DES MSC compared with ionic liquid-based and other 2D materials-based MSC.

voltage of 2 V. The CV patterns of CuTCPP-DES MSCs are symmetric, and DES as the electrolyte is quite stable, preventing the redox reactions of Cu^{2+} between electrodes and electrolyte (Fig. 3e; Fig. S12†). Therefore, the charge storage behavior of CuTCPP-DES MSC is mainly associated with the electric double-layer capacitive mechanism.²⁵

Complementing this, the GCD curves of CuTCPP-DES6 MSC recorded at $0.1\text{--}2 \text{ mA cm}^{-2}$ revealed minimal voltage-drop and symmetric characteristics at elevated charge-discharge current densities, signifying low equivalent series resistance and high coulombic efficiency (Fig. 3f). The EIS test corroborates the exceptional electrochemical performance of CuTCPP-DES MSC (Fig. 3g). The CuTCPP-DES6 MSC exhibited the smallest equivalent series resistance (ESR) value resistance (3.42Ω), which is consistent with the highest ionic conductivity of DES6 under CuTCPP confinement. We attribute this phenomenon to the ultrathin 2D MOF construction, which not only minimizes contact resistance but also substantially shortens ion diffusion paths, resulting in reduced electron resistance. This superior performance was maintained across different current densities. Even at a high current density of 2.0 mA cm^{-2} , the CuTCPP-DES6 MSC retained a capacitance

of 49.6 mF cm^{-2} (Fig. 3h). In fact, the areal capacitance of CuTCPP-DES6 MSC (81.3 mF cm^{-2} at 0.1 mA cm^{-2}) markedly outstripped the previously reported MOF-based flexible MSC (28.3 mF cm^{-2} at 0.2 mA cm^{-2}).⁴⁶ For comparative purposes, we fabricated a GO-DES6 MSC using identical steps as outlined in Fig. S11.† Fig. S12a† reveals the superior capacitance of the CuTCPP-DES6 MSC over the GO-DES6 MSC, corroborating the notion of improved ionic conductivity of DES under CuTCPP confinement. The charge carriers in our micro-supercapacitor system comprise protons, Ch^+ and Cl^- with some protons participating in hydrogen bonding within ChCl-BD DES and others existing as unbonded entities. These unbonded protons can form hydrogen bonds with oxygen-containing functional groups on the active carbon (AC) surface, catalyzing electrolyte ion transport and thereby boosting electrical conductivity. Our evaluation of the CuTCPP-DES MSC's cycling performance involved 10 000 charge-discharge tests (Fig. 3i and Fig. S13†). After 10 000 cycles at 0.2 mA cm^{-2} , CuTCPP-DES6 MSC sustained 70% of its initial capacitance, exhibiting a high coulombic efficiency (97%, Fig. 3i). The decline in cycling performance may be attributed to a solid electrolyte interface (SEI) layer during the initial cycles.⁴⁷ The

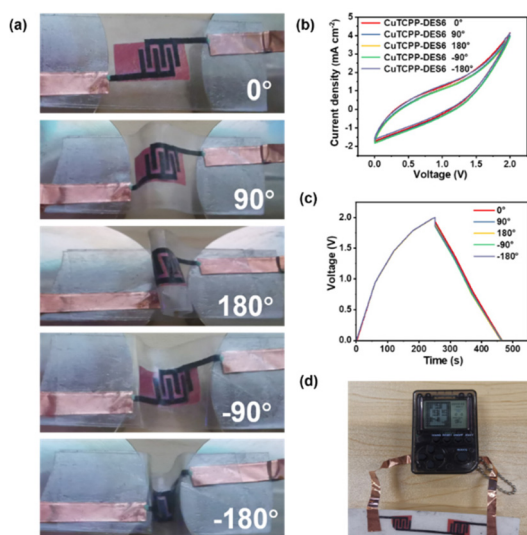


Fig. 4 (a) Photos of CuTCPP-DES6 MSC under different bending angles. (b) CV curves at 50 mV s^{-1} of CuTCPP-DES6 MSC bent at 0° , 90° , 180° , -90° , and -180° . (c) GCD curves at 0.1 mA cm^{-2} of CuTCPP-DES6 MSC bent at 0° , 90° , 180° , -90° , and -180° . (d) Photo of a game console powered by CuTCPP-DES6 MSC.

formation of this layer can consume active material and decrease the capacitance. However, once formed, this layer can stabilize the system and prevent further rapid decay and initially inactive material can start to participate in the charge storage processes. The observed minor fluctuations may be attributed to the residual stresses incurred during the fabrication process. The commendable capacitance and coulombic efficiency of the CuTCPP confined system could be ascribed to the higher ionic conductivity and lower transfer resistance, which together offer a more stable ion transport route and reduced energy loss. In terms of the energy storage, the CuTCPP-DES6 MSC produced an energy density of $45.17 \text{ } \mu\text{W h cm}^{-2}$ at a power density of 8.559 mW cm^{-2} , surpassing many similar ionic liquid-based MSCs and other 2D materials-based MSCs.^{6,32,48–54}

Mechanical flexibility is paramount for wearable and portable energy storage solutions. In the bending test, the flexible carbon cloth was used instead of the fragile carbon paper. The impressive mechanical flexibility of the CuTCPP-DES6 MSC under varying bending states of 0° , 90° , 180° , -90° , and -180° is shown in Fig. 4a. The CV and GCD curves revealed minimal alterations at these bending angles (Fig. 4b and c), indicating excellent mechanical flexibility and electrochemical stability. To illustrate its practicality, we connected a flexible CuTCPP-DES6 MSC to a handheld game console, which it successfully powered (Fig. 4d).

Conclusion

Among the varying CuTCPP-DES systems we examined, the ChCl:BD (1:6) nanoconfined system outperformed the others, demonstrating that an optimal molar ratio of the

hydrogen bond acceptor and donor is crucial for maximizing ionic conductivity. Such high ionic conductivity is attributed to the multi-ion transport channels brought by the porous nature of the MOF material. Leveraging this high ionic conductivity, we fabricated CuTCPP-DES6-based membranes as QSSEs to investigate the electrochemical performance of the nanoconfined system. The results revealed that CuTCPP-DES6 MSC presents the highest areal specific capacitance of 81.3 mF cm^{-2} at 0.1 mA cm^{-2} and an energy density of $45.17 \text{ } \mu\text{W h cm}^{-2}$ at a power density of 8.559 mW cm^{-2} , whose performance is superior to that of ionic liquid and 2D material-based MSC. Importantly, the performance remains impervious to bending, indicating its robust and versatile applicability. This wearable CuTCPP-DES MSC exhibits remarkable performance, hinting at extensive possibilities in the energy storage sector. The highly conductive, cost-effective, MOF material-based nanoconfined QSSE system developed in this study could greatly contribute to the advancement of electrolytes for MSCs and other electronic materials, thereby opening new directions in the field.

Author contributions

Xiaoyu Wang: Data curation, investigation, methodology, writing – original draft, writing – review and editing. Yuqi Wang: Conceptualization, supervision, writing – review and editing. Kang Yuan: Writing – review and editing. Bing Yao: Writing – review and editing. Xinsheng Peng: Conceptualization, funding acquisition, supervision, writing – review and editing.

Conflicts of interest

There are no conflicts to declare.

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